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Review

Novel community data in ecology-properties and prospects

Florian Hartig, [1,*] Nerea Abrego, [2] Alex Bush, [3] Jonathan M. Chase, [4] Gurutzeta Guillera-Arroita, [5] Mathew A. Leibold, [6] Otso Ovaskainen, [2,7] Loïc Pellissier, [8,9] Maximilian Pichler, [1] Giovanni Poggiato, [10] Laura Pollock, [11] Sara Si-Moussi, [10] Wilfried Thuiller, [10] Duarte S. Viana, [12] David I. Warton, [13] Damaris Zurell, [14] and Douglas W. Yu [15,16]

New technologies for monitoring biodiversity such as environmental (e)DNA, passive acoustic monitoring, and optical sensors promise to generate automated spatiotemporal community observations at unprecedented scales and resolutions. Here, we introduce 'novel community data' as an umbrella term for these data. We review the emerging field around novel community data, focusing on new ecological questions that could be addressed; the analytical tools available or needed to make best use of these data; and the potential implications of these developments for policy and conservation. We conclude that novel community data offer many opportunities to advance our understanding of fundamental ecological processes, including community assembly, biotic interactions, micro- and macroevolution, and overall ecosystem functioning.

Highlights

In recent years, new technologies have emerged that can generate rapid and standardized biodiversity inventories without explicit human guidance (novel community data).

The benefits as well as technical challenges of these technologies have been extensively reviewed, and ecologists are currently in the process of incorporating them into their observational studies.

So far, however, large novel community datasets are still rare. Consequently, there are still many open questions about how these new data should be optimally used to address fundamental questions in community ecology, macroecology, and conservation.

We review the state of the field, highlight the opportunities and analytical tools for advancing ecological research with novel community data, and discuss the implications of these emerging technologies for ecological theory, ecological study design, and environmental management.

Novel community data – introduction and definition

Understanding the factors that govern the distribution of Earth's biodiversity across space and time remains one of the most pressing problems in biodiversity science. While human activities are rapidly altering the structure of biodiversity and the services it provides to humans [1], our ability to describe, model, and manage these changes is hampered by the fact that conventional **biodiversity monitoring** (see Glossary) is limited in its spatial, temporal, and taxonomic scale and resolution, and is often poorly standardized and structured [2].

In recent years, major technological innovations in sensor technologies have occurred that promise to automate biodiversity monitoring. These include **eDNA**, **passive acoustic monitoring** [3–5], and **visual sensors** (e.g., camera traps, see [6]), which, coupled with appropriate machine learning or deep learning **pipelines** [7,8], are moving the field 'towards the fully automated monitoring of ecological communities' [9,10]. Hereafter, we refer to the **community inventories** generated by automated sensors and pipelines that do not directly involve humans in the detection and identification of species as **novel community data** (see also [11]).

The emergence of novel community data is likely to transform the way species distribution and abundance data are generated for the rest of the 21st century (e.g., [12–14]). The efficiency gains are such that hundreds or even thousands of species can be routinely detected and potentially quantified in their abundance across entire landscapes, resulting in a 'many-row, many-column' **community matrix**. These datasets are larger and richer in information than traditional community inventories, but they also have complicated properties such as higher rates of false positives or, in the case of eDNA, unreliable information on the relative abundance between species [15,16]. Novel community data therefore require appropriate statistical tools that can exploit their increased information content while also accounting for their added complications [17].

[1]Theoretical Ecology, University of Regensburg, Regensburg, Germany
[2]Department of Biological and Environmental Science, University of Jyväskylä, P.O. Box 35 (Survontie 9C), FI-40014 Jyväskylä, Finland
[3]Lancaster Environment Centre, Lancaster University, Lancaster, UK
[4]German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig, Leipzig, Germany
[5]Pyrenean Institute of Ecology, CSIC, Jaca, Spain
[6]Biology Department, University of Florida, Gainesville, FL, USA

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The sensors and technologies used to generate novel community data have been extensively reviewed elsewhere [9,11,12,18–24]. In this review, we will therefore only briefly cover this topic and focus instead on how the combination of novel community data with new statistical tools both compels and enables us to transform data analysis, expand our scientific reach, and improve the conservation and management of biodiversity.

What makes novel community data really novel?

Over the past two decades, ecologists have assembled large collections of spatial occurrence or abundance observations [e.g., Global Biodiversity Information Facility (GBIF), International Union for Conservation of Nature (IUCN) range maps, or taxa-specific monitoring schemes]. These data are frequently used in **species distribution models** (**SDMs**, e.g., [25,26]) to estimate species' environmental niches, project future distributions under climate or land-use change, or generate biodiversity metrics for conservation and management. A commonly recognized limitation of these data, especially when they are opportunistically collected, is uncertainty about observation errors and intensities [27]. Moreover, these data are rarely suitable for inferring local community co-occurrences across trophic groups, limiting their potential for understanding the role of **biotic interactions** in community and ecosystem dynamics.

Dedicated conventional data collection schemes exist that provide both the presence and (somewhat reliable) absence, or abundance/biomass information for entire local communities across space [28]. However, using conventional survey techniques, such data are typically limited in their sample size, spatial and temporal extent, and especially in taxonomic coverage and resolution (see [20], but see [29]).

The emergence of novel community data (Figure 1) promises to fundamentally alter this established landscape of biodiversity observations. It is tempting to dismiss our ability to sequence eDNA, ancient DNA, and bulk-sample DNA [20,21,24,30] (Box 1), as well as the availability of camera traps or passive acoustic monitoring, as merely a convenient way to generate more data (i.e., big data) of the same kind that we have been collecting. Such a view, however, neglects the many other dimensions in which novel community data differ from traditional community inventories.

Structure and standardization

Especially as technology evolves and pipelines are shared, compared, and converge on common standards, novel community datasets have the potential to be more structured and standardized than traditional sampling schemes. Moreover, novel community data are typically generated according to a fixed plan using low-expertise collection methods, positive and negative controls, and a standardized processing pipeline for species identification. Therefore, results are less dependent on individual observers.

Importantly, by standardized, we do not mean error-free. For example, eDNA data can have considerable errors (Table 1). However, because these errors are usually more consistent and are therefore somewhat predictable, they can be more easily corrected using statistical methods, relative to errors in conventional surveys that arise from different human observers or subtle differences in sampling protocols.

Spatial, temporal, and taxonomic extent and resolution

A second difference is that the automated way in which novel community data are generated makes them scalable to high spatial, temporal, and taxonomic resolution [30,31]. Different sensors have different strengths in these dimensions, but these can be combined by using multiple

[7]Organismal and Evolutionary Biology Research Programme, Faculty of Biological and Environmental Sciences, University of Helsinki, P.O. Box 65, Helsinki 00014, Finland
[8]Ecosystems and Landscape Evolution, Institute of Terrestrial Ecosystems, Department of Environmental Systems Science, ETH Zürich, 8092 Zurich, Switzerland
[9]Unit of Land Change Science, Swiss Federal Research Institute for Forest, Snow and Landscape Research (WSL), 8903 Birmensdorf, Switzerland
[10]Univ. Grenoble Alpes, Univ. Savoie Mont Blanc, CNRS, LECA, F38000, Grenoble, France
[11]Department of Biology, McGill University, Montreal, Quebec, Canada
[12]Doñana Biological Station (EBD-CSIC), Seville, Spain
[13]University of New South Wales, Sydney, Australia
[14]University of Potsdam, Potsdam, Germany
[15]Kunming Institute of Zoology; Yunnan, China
[16]University of East Anglia, Norfolk, UK

*Correspondence:
florian.hartig@ur.de (F. Hartig).

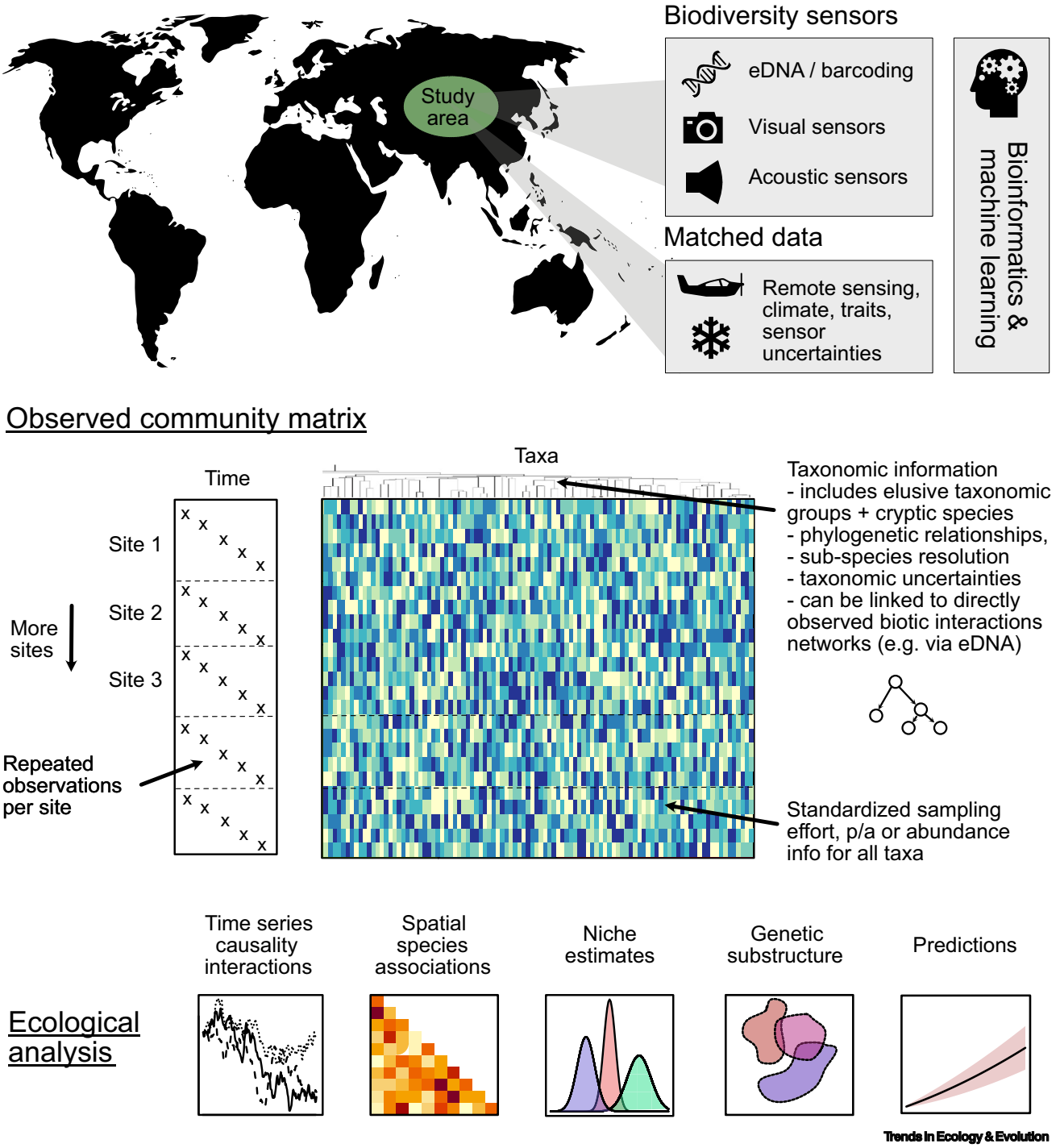


Figure 1. Novel biodiversity sensors generate detailed community inventories as well as rich metadata. If replicated in space and time, this gives rise to novel community data. This novel community, represented in the center of the figure, is more information-dense in many dimensions beyond spatial replicates, including time, taxonomic relationships, and interaction information. As a result, these data allow for a richer set of ecological analyses than conventional community inventories. Abbreviations: eDNA, environmental DNA; p/a, presence–absence.

sensor types (see also [14]). For example, while eDNA data have particular strengths in taxonomic breadth and resolution, as well as detection sensitivity and hence community completeness (Box 1), acoustic and visual sensors are better at producing continuous community time series. Indeed, acoustic and visual sensors offer the unique opportunity to continuously capture biodiversity over daily, seasonal, and even decadal time scales, something that is difficult to achieve with nonautomated sampling schemes. An obvious advance for the field would be to use statistical methods to combine observations from these different data streams into a combined spatiotemporal data product or model (cf. [20,32]) (see Outstanding questions).

All sensors can, in principle, also be used to estimate abundance, although this will typically require additional steps (for eDNA, [33] and Box 1). Next-generation methods may even allow individual-level identification and tracking (via genetic data or image analysis) to investigate behavior, dispersal, or migration patterns. Moreover, with eDNA, we can also identify taxonomic patterns below and beyond the species level, such as **exact-sequence variants (ESVs)** or genetic diversity within and between species [34].

Metadata acquisition and matching to other data sources

Another advantage of using standardized sensors, rather than humans, is that **metadata** can be easily recorded during data acquisition. Metadata typically include time and location stamps, and, importantly, instrument errors and taxonomic uncertainties, which are rarely recorded in conventional surveys. Universally available metadata on time, location, and taxonomy facilitate matching observations to other local sensors and independent data products, such as weather stations, remote sensing data, phylogenetic or trait information, or biotic interactions extracted from visual, acoustic data, or eDNA analysis [35]. The resulting combined data products could be of interest as essential biodiversity variables for the GEO-BON platform [36]. We acknowledge that the collection of rich metadata is considered best practice for conventional biodiversity inventories as well; however, we believe that in practice, automated sensors are likely to collect richer and more structured metadata than conventional surveys.

Observation errors and data quality

Despite these advantages, ecologists are often skeptical about the quality and reliability of novel community data. We recognize that each sensor type presents certain technical challenges, some of which are inherent in the measurement process (e.g., the field of view of a camera) and others in the analysis pipeline (e.g., for eDNA, incomplete **DNA barcoding** reference databases or PCR errors; for acoustic and visual sensors, transferability of deep learning methods for species recognition). The two-step process of the measurement itself and the pipeline of analysis and species identification can introduce errors and biases that are more complex than conventional data (Box 1 contains a discussion of the eDNA pipeline). However, the development towards standardized pipelines and protocols, as well as the collection of rich metadata, also offers many opportunities to account for such errors in subsequent statistical analyses (see the section 'Statistical models to deal with observation errors').

Using novel community data to answer long-standing ecological questions

Having established that novel community data will provide not only a larger sample size, but also a richer, more standardized, and more interconnected data product than traditional biodiversity monitoring data, we focus on how these data will transform the way we can approach classical and new ecological questions. We organize this discussion around five themes: (i) **species associations**; (ii) biotic, especially trophic interactions; (iii) beyond the species concept; (iv) real-time monitoring and long time series; and (v) understanding ecosystems as complex systems.

Glossary

Biotic interaction: a direct (e.g., competitive, mutualistic, or trophic) interaction between individuals of two different taxa.
Biodiversity monitoring: the process of generating information about the spatiotemporal distribution of biodiversity. The data thus produced are often represented as a community matrix (see later).
Community inventory: a list of species occurring in a particular place and time, also referred to as a biodiversity inventory. Conventional inventories often target a particular species group.
Community matrix: a matrix consisting of many community inventories, where the rows traditionally indicate the inventory number (sites or time) and the columns indicate species or taxa, thus characterizing the presence, presence–absence, abundance, or biomass for each species–site combination.
Cryptic species: species that are morphologically indistinguishable but genetically distinct and reproductively isolated and can thus only reliably be identified by molecular analyses.
DNA barcoding: identification of a species using a short section of DNA from a specific gene or genes, which is mapped against a barcoding reference database.
Environmental DNA (eDNA): DNA isolated from environmental samples, including both extraorganismal (trace) and organismal eDNA. For example, bulk-arthropod samples contain both organismal eDNA from arthropods and trace eDNA from vertebrates (e.g., blood, feces, and skin).
Exact-sequence variants (ESVs): unique DNA sequences that are identified by high-throughput sequencing. Unlike more traditional operational taxonomic units (OTUs, see later), which cluster nonidentical but similar sequences, ESVs describe identical nucleotide sequences.
Joint species distribution model (jSDM): a statistical model that describes a vector of community (multispecies) presence or abundance as a function of abiotic, biotic, or spatial predictors (similar to an SDM) and an additional component, which consists of residual covariances between the

Species associations

Because novel community data can provide complete community inventories, they are well suited for investigating species associations. Raw species associations can arise from shared environmental preferences, but even when these are accounted for (see the section 'Statistical tools for novel community data'), species still often show associations. These associations may be artifacts caused by unmeasured or inadequately measured environmental or spatial factors (e.g., [37–39]), but they may also reflect biotic interactions. The ability to comprehensively quantify species associations, especially when used in conjunction with direct observations of biotic interactions (see the next subsection), offers the potential to advance the long-standing goal of disentangling spatial, abiotic, and biotic factors as drivers of (meta)community assembly [40–42]. Moreover, if the data contain both spatial and temporal dimensions, associations can be investigated over both time and space, which may be critical to infer the underlying processes of metacommunity assembly [43]. Finally, even if the causes of spatial associations cannot be resolved, they reduce unexplained variation in the community composition and thus may provide a more realistic estimate of the irreducible stochasticity in community dynamics and assembly rules (e.g., [41]).

Biotic interactions

Novel community data, particularly eDNA data, can also be used to directly infer species' interactions, both trophic and mutualistic [44]. The most straightforward way to observe trophic interactions and thus infer entire food webs is to sequence the gut contents of individuals (see, e.g., [45], who sequenced the gut contents of coral reef fish to reconstruct a complex marine food web). It is also possible to infer host–vector–pathogen networks [46] or mutualistic interaction networks from interaction residues, for example, by analyzing pollen on pollinators [47] or eDNA traces on flowers [35]. Such direct observations of species' interactions can be compared with species' associations or data on disturbances (e.g., [48]) to understand how these biotic interactions affect community assembly, ecosystem dynamics, or species distributions.

Beyond the species concept

Another area where eDNA data in particular could lead to advances is in challenging the near-exclusive role of species as the basic unit for quantifying biodiversity and community patterns. While we believe that the species concept will remain central to ecology, novel community data can increase taxonomic resolution to the subspecies or even ESV level. This would not only solve the problem of **cryptic species** [49] but could also reveal large-scale 'macrogenetic' patterns of interspecific genetic variation and gene flow (cf. [50,51]). An important question is how a more 'granular' view of a species' distribution could be integrated into concepts such as competition, distribution, the niche, or extinctions, which are central to both ecology and practical conservation (e.g., [52,53]).

Real-time monitoring, nowcasting, and ancient DNA

A natural advantage of acoustic and visual sensors over eDNA is their high temporal resolution, which offers the potential to observe short-term changes in population size, species interactions or habitat preferences, or phenological changes, as well as community time series (e.g., [21], Figure 2). This offers the potential for real-time monitoring and nowcasting of biodiversity changes, biological invasions, and pathogen outbreaks [54,55]. Another interesting idea is the ability to generate observations and time series from the past using ancient DNA [21,56], which could be critical for understanding human impacts on ecosystems in the Anthropocene.

Ecosystems as complex systems

Finally, the fact that novel community data provide direct measurements of species interactions (i.e., the trophic structure) together with community inventories at high spatiotemporal resolution

modeled species, describing positive or negative associations.
Metadata: in general, data describing other data. In the context of this paper, we include all data that complement the primary observations of the community in this definition.
Novel community data: large community datasets generated by automated pipelines such as eDNA sequencing and electronic sensors (e.g., bioacoustic sensors or visual sensors such as camera traps).
Operational taxonomic unit (OTU): a group of haplotypes that are clustered together on the basis of their sequence similarity to form distinct taxonomic entities, typically species.
Passive acoustic monitoring: deployment of acoustic sensors in the field to detect sounds created by wildlife and the surrounding (soundscape). These data can be processed by experts or machine learning methods to classify the sounds of specific species or communities.
Pipeline: a series of computational and analytical steps to process and analyze raw sensor data such as sequencing data, acoustic observations, or pictures.
Species distribution model (SDM): a statistical model that relates species presence or abundance data to a set of abiotic, biotic, or spatial predictors.
Species association: a correlation or association of occurrence, abundance, or distribution of two taxa, which can be due to biotic interactions, (missing) environmental covariates, distributional disequilibrium, and other reasons.
Visual sensors: we use visual sensors as an umbrella term for all optical sensors that can be used for species identification. These include photos (e.g., from camera traps), videos, and potentially also visual information from remote sensing, in particularly from drones.

Box 1. An overview of the eDNA pipeline

All species shed DNA into the environment. We refer to this DNA isolated from environmental substrates, even the air [82,83], as eDNA [24,84,85]. eDNA can either be sequenced *en masse* and processed *in silico* to find taxonomically informative sequences (metagenomics), or read after targeted amplification of taxonomically informative sequences in the laboratory (metabarcoding). The resulting DNA sequences ('reads') are typically first clustered to **operational taxonomic units (OTUs)** and then compared with DNA barcode reference databases to assign taxonomies [86].

Although the eDNA pipeline can, in principle, detect all cellular organisms, the taxonomic coverage achieved in current eDNA studies is limited by the physical collection of eDNA material, by the molecular methods used, and, for taxonomic assignment beyond OTUs or ESVs, by the availability of suitable reference databases [87]. Future methods are likely to expand the taxonomic coverage, but even existing methods enable the standardized detection of many species across trophic groups, including cryptic, difficult-to-observe, small, and less abundant species, from easily collected samples.

Practical challenges in using eDNA include the high diversity of different bioinformatic pipelines for curating, cleaning, and clustering the eDNA sequences (but see [88]), as well as dealing with eDNA-specific sampling and detection errors (see Table 1 in main text, see also [15,75,89,90]). For example, stochasticity and sample-equalization steps in laboratory pipelines can obscure the expected positive relationship between the biomass of eDNA and the resulting number of reads, but adding a DNA spike-in to each sample can help to recover this relationship [75,90]. Moreover, sample contamination can result in false positive errors. Good practice limits such events to being rare and weak, letting false positives be identified [91].

A further challenge with eDNA data is that the number of eDNA reads per individual depends, in part, on unknown species-specific rates of release, degradation, and PCR efficiency ('species effects', see Table 1 in main text) (see also [16]). As a result, eDNA reads are, in general, not proportional to species' abundance or biomass. However, if (i) eDNA release, degradation, and PCR efficiency are approximately constant across samples, and (ii) pipeline stochasticity is accounted for (via spike-in estimated offsets), then cross-sample changes in reads for each species are proportional to cross-sample changes in that species' abundance [33,75,90,92].

Finally, taxonomic assignment can have errors or uncertainties resulting from incomplete reference databases and variation across species in their genetic diversity. Ideally, such errors are accounted for by dedicated statistical methods. For example, Bayesian algorithms can be trained to estimate the degree of sequence similarity required to assign membership to a given rank within a given taxon [93,94].

may help us to revive the old aspiration of 'modelling all life on Earth' [57], that is, understanding ecosystems holistically as complex systems and describing their various interactions through mechanistic ecosystem or macroevolutionary models (e.g., [58]).

Statistical tools for novel community data

The 'law of the instrument' famously warns us that 'if all you have is a hammer, everything looks like a nail'. The saying cautions us that instruments and analytical tools, rather than scientific curiosity, often determine what research questions are asked. While the availability of new sensors expands our toolbox for data collection, tailored analytical approaches for novel community data are still rare, which currently limits our ability to use these data for answering the ecological questions we listed in the previous section. We see three main directions in which statistical methods for novel community data should be developed: community and metacommunity analysis, time series analysis, and network analysis.

Community and metacommunity analysis

Community and metacommunity analysis aim to understand how community composition changes as a function of the environment and possibly interactions between communities. Statistically, we can approach this problem from at least three angles: we can use differences or changes in community composition as a response (e.g., ordination, Mantel tests, or regressions on distance matrices [59]); we can use constrained ordinations to partition effects on community composition between spatial and environmental predictors; or we can develop statistical models that predict community composition directly [as achieved, e.g., in **joint species distribution models (jSDMs)**, see [60–62] and Box 2]. While each of these approaches has its strengths,

Table 1. The two stages of DNA-based surveys and the sources of false negative errors; false positive errors; and row, column, and cell effects in the output sample x species table (adapted from [75])

Stage 1: eDNA biomass collection		Analogs in conventional surveys
Species effects	Every sample collects a certain amount of eDNA biomass of each species, which is proportional to the species' biomass available at the site. However, the proportionality constant is marker- and species-specific and is unknown, since rates of DNA release, 'catchability', and degradation differ across species and physiological states (a 'column' effect).	Species differ in their detectability by human observers or by trapping bias.
Noise	The amount of eDNA biomass collected per species varies stochastically among samples collected at the same site and time (a 'row' effect), including outright collection failure (false negatives).	Imperfect detection of species, false negatives
Error	It is possible for traces of eDNA from elsewhere to contaminate a sample (false positives).	No analog in conventional surveys
Stage 2: eDNA laboratory + bioinformatics pipeline		Analogs in conventional surveys
Species effects	Species differ in extraction efficiency, gene copy number, and PCR amplification efficiency, causing the relationship between the amount of input eDNA and number of output sequence reads to be species-specific (a 'column' effect).	Species differ in their detectability by human observers or by trapping probabilities.
Pipeline effect	PCR stochasticity, normalization steps, and the passing of small aliquots of liquid along the laboratory pipeline add stochasticity to the total number of reads output per replicate of the sample (a 'row' effect), including outright detection failure (false negatives).	No analog in conventional surveys
Noise	On top of species and pipeline effects, there is additional noise in the number of reads per species, sample, and/or technical replicate (a 'cell' effect).	No analog in conventional surveys
Contamination Error	It is possible for traces of eDNA from one sample to contaminate other samples (false positives).	No analog in conventional surveys
Barcoding errors	Incorrect delimitation of sequence variation leading to incorrect taxonomic lumping or splitting; or incorrect identification of a species because the sequence is wrongly assigned to a taxonomy (paired false negative and false positive errors)	Incorrect lumping of cryptic species or incorrect splitting of a single species; or misidentification of a species resulting in paired false negative and false positive errors

we find the option of modeling communities directly with jSDMs particularly promising because it allows us to infer species-specific environmental preferences, spatial effects, and species associations, all of which are quantities that are biologically interpretable and are useful for making predictions.

Time series for inferring causal drivers

Apart from a few exceptions, conventional monitoring has so far been unable to provide continuous time series over large spatial scales and long periods of time. This is unfortunate, because time series are better suited than static data for separating correlation from causation. A prominent idea in causal time series analysis is the concept of Granger causality [63], which posits that because the cause must precede the effect, we can regress our observations (in this case, the community composition at each time step) against the observations of previous time steps. This approach could also be used to infer asymmetric interactions (and thereby hierarchical competition), and it has been argued that interactions based on such a temporal or spatiotemporal approach are more likely to match true biotic interactions (see [64] and Figure 2, for an implementation in an extended jSDM). Novel community data, especially acoustic and visual sensors, can provide continuous time series data at unprecedented rates. Therefore, we believe that these data could

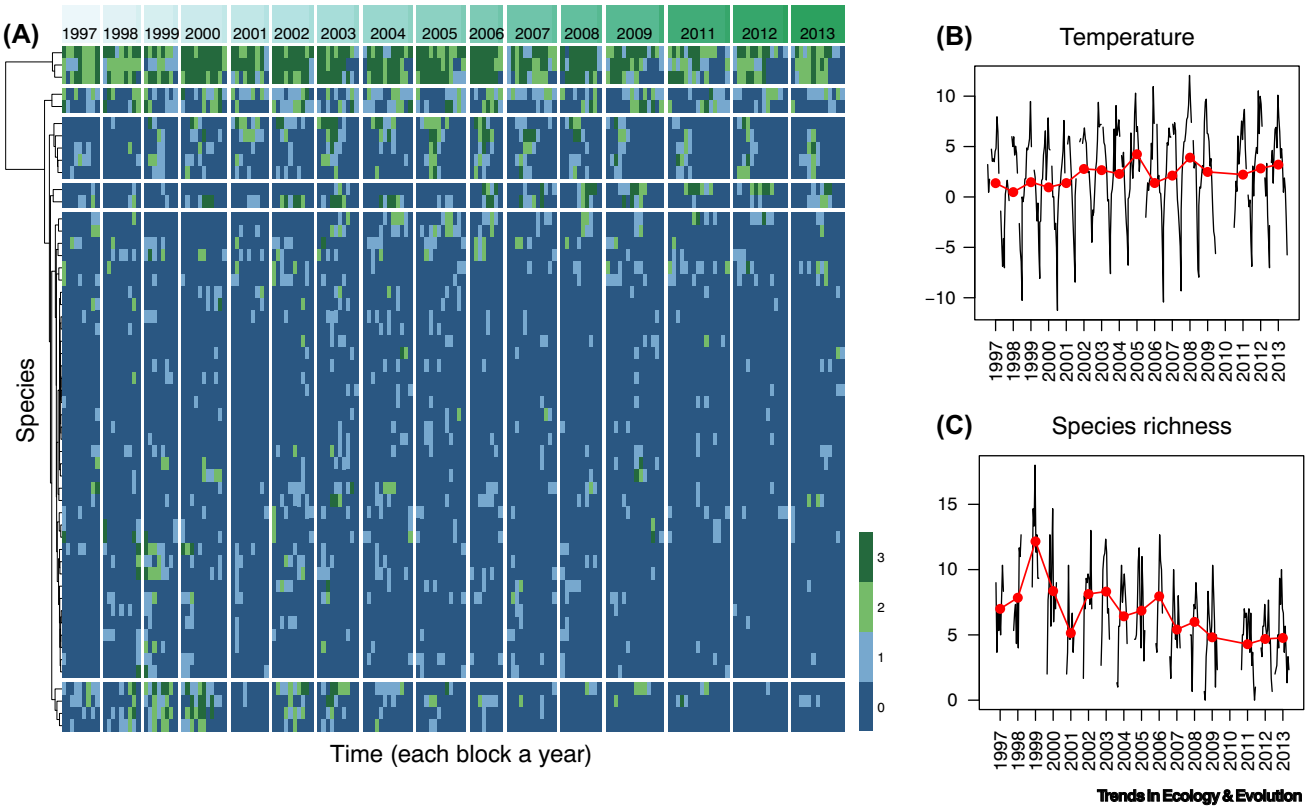


Figure 2. Abrego *et al.* [30] analyzed a 16-year weekly community time series of an arthropod community's dynamics in Greenland, resolved to the species level by environmental (e)DNA mitogenome mapping. Panel (A) shows the species x time community matrix, with cell colors indicating the number of traps out of three in which the species was detected at each point in time. During the study period, the temperature increased by 2°C (B) and the richness of arthropod species halved (C). Reprinted from [30]. In their analysis of the data, Abrego *et al.* showed that abiotic variables alone were insufficient to predict species responses, but when the interactions of the species were included, the predictive power of the model improved. Trophic cascades thereby emerged as being important in structuring the response of biodiversity to climate change. The study emphasized the potential of eDNA data to generate high-resolution community time series and thus to an understanding of the complex interplay of biotic and abiotic effects in the impacts of climate change. The analytical tools used to reach these conclusions are explained in Box 2.

be instrumental in inferring causal relationships among species or groups of species and in better understanding community assembly as a whole.

Network analysis

A third avenue for statistical analysis is to analyze and compare species association networks inferred through jSDMs, and networks of mutualistic, trophic, or competitive biotic interaction networks that are generated, for example, by sequencing gut contents (see also Figure 1). This line of research could leverage methods from the field of network analysis [65], which often struggles with the same data limitations as community ecology. Novel community data could allow us to analyze larger and more complex networks (e.g., [66]), analyze how these networks change across environmental gradients [67], and link these patterns to community data to understand how biotic interactions, in conjunction with environment and space, give rise to spatiotemporal biodiversity patterns [68]. For example, it has been found that species associations change with scale [69], but it is unclear whether such changes reflect anything about their underlying biotic interactions. Another example is that although two species interact locally (e.g., predator–prey), they may not show any association [70]. Understanding the interplay between association and interaction networks may be key to understanding the role of biotic interactions in structuring communities and spatial biodiversity patterns.

Statistical models to deal with observation errors

When designing these and other statistical analyses for novel community data, it is likely critical to incorporate observation models that account for detection probabilities and taxonomic uncertainties. Observation models are not specific to novel community data, but detection errors may be more pronounced and complicated in novel community data (e.g., Box 1). On the positive side, as a result of standardized pipelines and rich metadata, the errors and uncertainties in detection and taxonomic assignment may be easier to estimate. Currently, statistical models are emerging that correct species detections for false positives and negatives (e.g., [71,72]) and extend these ideas to communities and jSDMs [73,74], relative biomass estimates [75], and continuous-score observations [76]. A challenge for the future is to make these models more broadly accessible and ready for the computational demands of large novel community datasets.

Improving predictions of biodiversity responses to global change

Finally, novel community data could help to improve predictions of biodiversity dynamics under global or climate change beyond the trivial fact that more data are always useful. For example, spatiotemporal community data are better suited to identify causal effects and directional interactions ([63]; see also the section 'Time series for inferring causal drivers'). Identifying these factors is particularly important when predicting species or biodiversity responses outside present climatic conditions.

Leveraging novel community data to achieve socioecological resilience

Beyond scientific progress, novel community data may also enhance society's ability to create effective governance of biodiversity as a public good. In their seminal paper, Dietz *et al*. [77] described five elements for the successful governance of public goods: (i) information generation, (ii) infrastructure provision, (iii) political bargaining, (iv) enforcement, and (v) institutional redesign.

The most obvious role for novel community data is to contribute to the first element: the generation of high-quality, granular, and timely information on ecosystem status, health and change, uncertainty levels, values, and the magnitude and direction of anthropogenic impacts. In addition,

Box 2. Joint species distribution models (jSDMs) as a tool to model novel community data

In recent years, jSDMs have emerged as the main extension of classical species distribution models (SDMs), for the analysis of community data [60–62]. The key difference between SDMs and jSDMs is that while the former can also model communities, they do so by describing each species individually (stacked SDMs).

A jSDM, however, is a true community model because, additional to the environmental responses of each species, it includes a species–species covariance term. This covariance term models the associations of species, meaning the tendency of species pairs to co-occur more or less frequently than one would expect on the basis of their species-specific environmental preferences alone (Figure I).

The basic jSDM structure can be extended to include additional correlations in species' niche estimates via phylogeny or traits, and spatial predictors. jSDMs can also be extended to fit spatiotemporal data, which allows one to consider additionally asymmetric associations [63,64]. Due to their complex likelihood, jSDMs are often challenging to fit, and several numeric strategies, including latent variable approximation (e.g., [60]) and Monte Carlo approximations [95], have been proposed to make these models scalable to large community data.

The interpretation of the species' associations inferred by jSDMs has been the subject of considerable debate in the field. We view it now as accepted that species' associations are not necessarily caused by biotic interactions (e.g., [38], but see [37]). Among other things, this implies that a jSDM will typically not improve the estimation of the fundamental niche [39]. Nevertheless, the ability to partition the community signal into the three classical components of environment, space, and association (Figure IE), which can further be broken down to sites (communities) and species (i.e., the 'internal structure', see [41] and Figure IF), provides a rich framework for analyzing spatial community data. Moreover, if some species can be easily observed, conditioning on their presence using jSDMs can also improve predictions [96], which may be relevant for management.

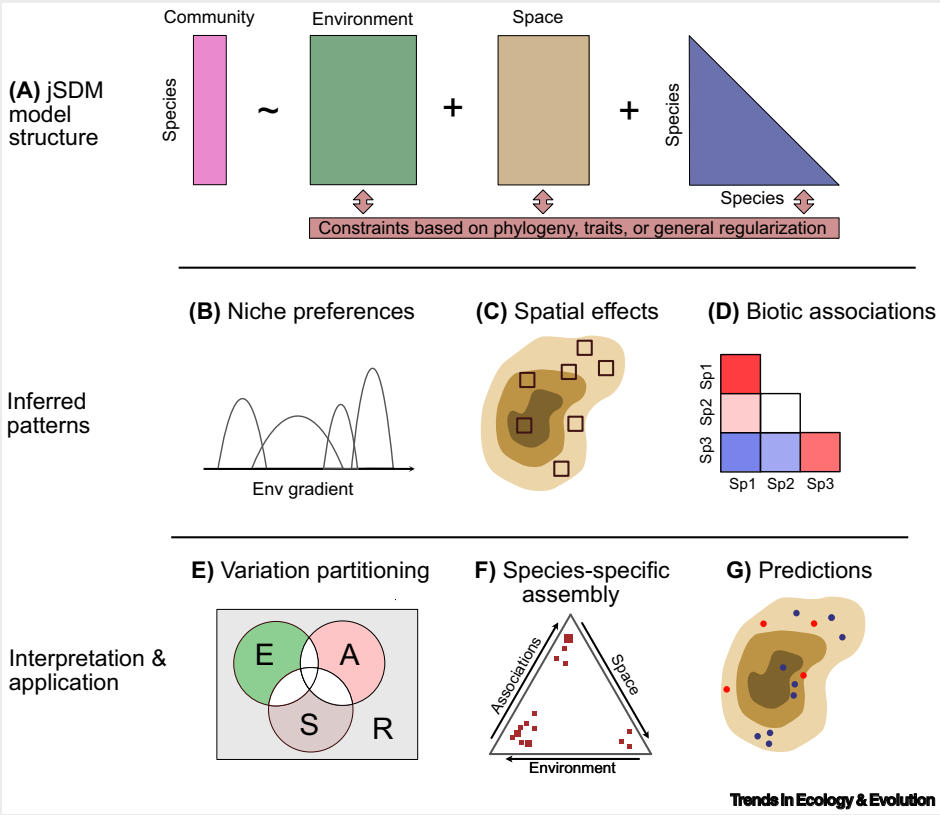


Figure I. An overview of structure, inferred patterns, and interpretation of a joint species distribution model (jSDM). (A) A possible jSDM structure, predicting community composition based on environment, space, and species–species covariance. (B) Environmental effects show niche preferences. (C) Spatial effects show spatial clustering of species. (D) Species–species covariance shows species associations. (E) An analysis of variance of the entire jSDM shown in (A) can partition community variation into the environment, space, associations, and residual components. (F) This can further be broken down by species or sites [41], so that we can see the relative importance of the three components to individual species and sites. (G) If particular presences are known (red), we can condition on them to improve predictions [96].

as new infrastructure allows methods to become more automated, independent parties can collect, analyze, and compare large biodiversity datasets, making this knowledge more understandable and trustworthy [78]. Information with these properties can, in turn, make political bargains more achievable and enforcement more effective. Governments could apply 'technology forcing' to encourage the creation of novel community data [79] and, ultimately, redesign environmental institutions for greater effectiveness, as exemplified by the UK's Great Crested Newt offset market (Box 3).

Moreover, novel community data could also provide opportunities to redesign scientific and political structures. For instance, although most regulatory uses of eDNA still involve only single-species detection [79], in the USA, these data are being combined into a multispecies database, the Aquatic eDNAtlas Project. To facilitate such a process, rigorous sampling protocols, reference datasets and pipelines for creating biodiversity data [e.g., artificial intelligence (AI) models for species recognition, barcode databases) should be applied that are freely available and integrated into global monitoring schemes and databases such as GBIF, IUCN, and GEOBON (e.g.,

Box 3. An eDNA-enabled biodiversity offset market

One example of institutional redesign enabled by eDNA is the district licensing market for the great crested newt (*Triturus cristatus*), a protected species in the UK. Developers are required to survey for the newt when their plans may affect ponds, and to respond to newt detections by paying for mitigation measures. Traditional surveys require at least four visits per pond during the short breeding season, using multiple methods that are only effective at night. Following a study [97] showing that a single eDNA water survey could detect the newt with the same sensitivity as traditional surveys (i.e., eDNA detection is high-quality and granular), the government authorized newt eDNA surveys in 2014, and a private market for eDNA surveys, audited with proficiency tests, grew to provide the infrastructure for timely and trustworthy information [98].

The switch to eDNA surveys increased survey efficiency, but the UK's reactive (mitigate after impact) approach was initially left in place. Mitigation measures, such as translocation, can take over a year, with associated costs. In 2018, the UK government took further advantage of eDNA's efficiency by implementing an institutional redesign with the district licensing scheme, in which the ponds across one or more local planning authorities are systematically surveyed with eDNA [99]. The data are then used to fit a model of the species' distribution, which is made into an understandable map of discrete background risk zones for the newt (Figure I). Builders can meet their legal obligations at any time by paying for a license, the cost of which depends on the size of their site, the background risk zone, and the number of ponds affected.

The fees from these licenses are mainly used for the proactive creation and long-term management of compensatory habitats, including ponds with a 1:4 impact-to-gain ratio. The compensatory habitat is directed towards Strategic Opportunity Areas that account for the planning authority's building aspirations (political bargaining). Enforcement takes place through the same processes that apply to all planning permissions. Both the UK government and a private–public– non-governmental organization (NGO) partnership run versions of district licensing markets, which, together, have reported creating hundreds of new ponds and associated habitat. In the future, it might be possible to effect a further institutional redesign by exploiting the multispecies information in the pond water samples to move to multispecies conservation planning and offset markets [100].

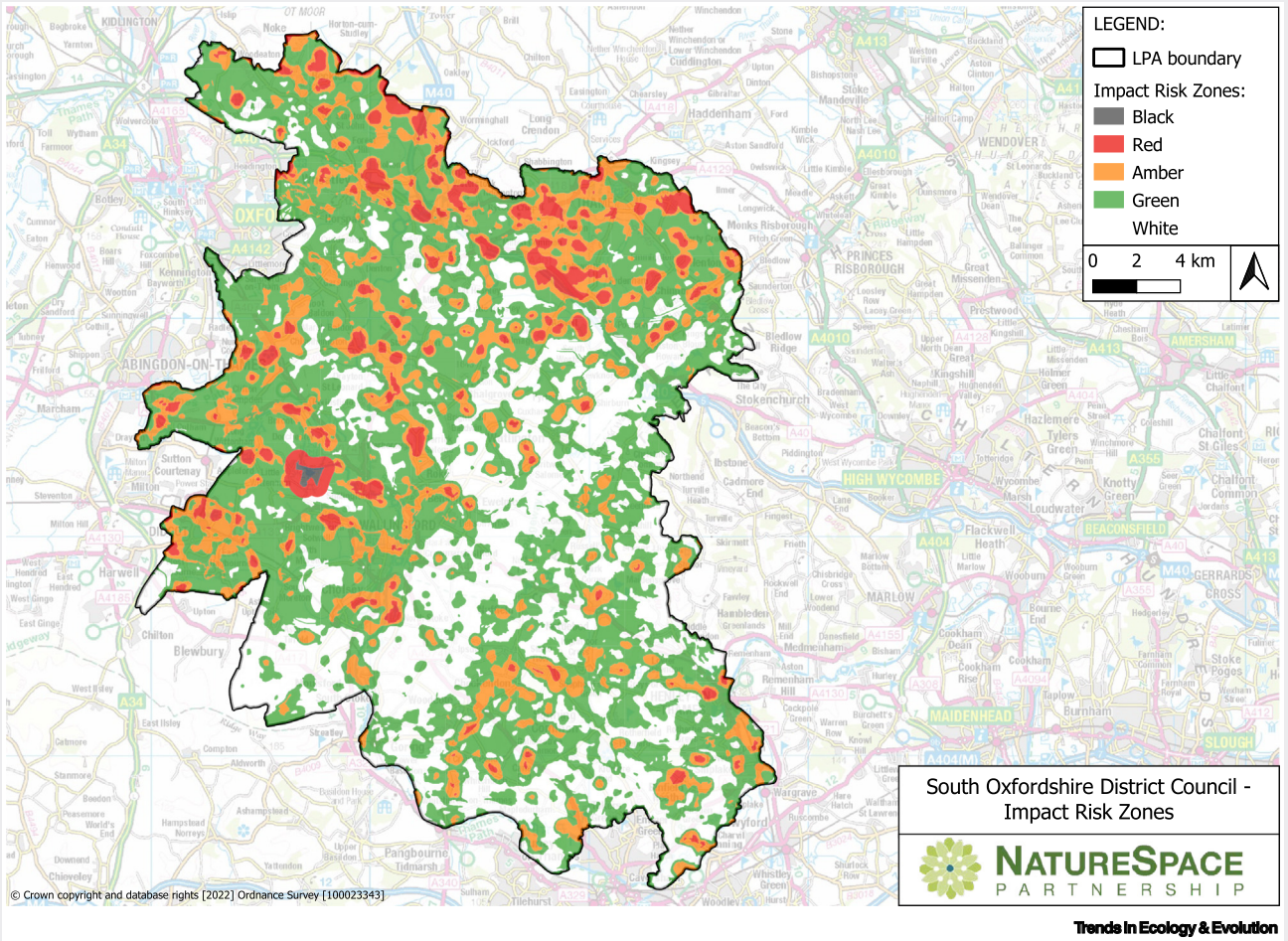


Figure I. Risk zone map for great crested newt (*Triturus cristatus*) in one local planning authority (LPA). Reprinted with permission from NatureSpace Partnership.

[22,80]). Based on these, policy-relevant data products such as global biodiversity integrity maps with granular and timely data (e.g., STAR, see [81]) could be created. Bayesian optimal design could be used to identify data gaps and thus to prioritize funding for initiatives to fill these gaps. For industry, the availability of such data can help to integrate ecological impacts into corporate decision-making. For example, the Taskforce on Nature-related Financial Disclosures (TNFD, https://tnfd.global/) has developed an analytical framework for assessing corporate exposure to nature-related risks and opportunities.

Concluding remarks and future perspectives

Novel community data offer exciting opportunities for understanding and predicting biodiversity patterns. For the first time, we can hope to generate spatiotemporal community inventories with high spatial, temporal, and taxonomic resolution, in conjunction with traits, abiotic predictors, and observed true biotic (mutualistic and trophic) interactions. While the need for and value of such multifaceted biodiversity data has been acknowledged for some time, the emergence of sensors that inherently produce community-level rather than single-species data at scale have brought the achievement of this long-held goal within our immediate reach.

The lower cost, more complex structure, and higher information density of these data have important implications for how we can conduct and advance ecological analyses, concepts, and theories. We have argued that (joint) species distribution models, network analysis, and time series, paired with statistical tools inherited from causal analysis, could serve as some of the core analytical tools to connect these data to important ecological research questions, particularly in niche theory, metacommunity theory, and network theory. Beyond this, novel community data also have great potential to provide crucial information for environmental management and biodiversity conservation.

Challenges for the future include the creation of appropriate data products, which would include establishing standardized field designs and pipelines, and bringing together existing data in common databases; the establishment of accessible statistical models to analyze these data, and the use of these analytical tools to produce ecological theories as well as actionable predictions for management and conservation.

Outstanding questions

How can we combine novel community data from different sensors to best characterize biodiversity patterns, multitrophic networks, and ecosystem dynamics?

How can observation models deal with the specific errors and idiosyncrasies of the different sensors?

What can we learn about communities by taking a more 'granular' taxonomic approach, looking beyond species as the main unit of taxonomic classification?

What analytical methods are best suited to exploit the properties of novel community data, particularly the extended taxonomic breadth and resolution, time series, and rich metadata?

How can these methods be linked to ecological concepts and theories in macro- and meta-community ecology, including niche theory and community assembly theory?

How can novel community data be used to predict biodiversity responses to global change?

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Declaration of interests

D.W.Y. is a cofounder of NatureMetrics, which provides commercial eDNA services and is a minor shareholder in the NatureSpace Partnership. No further conflicts of interest to declare.

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